

Assignment - I

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1. Explain the concept of system services provided by an operating system.

→ System services provided by an operating system.

- System services are the facilities by an operating system (OS) to make it easier for users and application programs to use the computer system. They act as a bridge between applications / users and computer hardware.

1. program execution :-

The O.S loads program into memory, runs them and handles their termination.

2. I/O operations :-

It provides a standard way for programs to perform input and output operations, such as x

3. File System Manipulation :-

The O.S allows users and programs to create, delete, read, write, rename

and organize files and directories.

4. Communication:-

The O.S provides mechanism for processes to communicate with each other, either on the same computer or across a network.

5. Error detection:-

The O.S detects errors in hardware, memory, I/O devices and running programs and takes appropriate action to maintain system stability.

6. Resource Allocation:-

Since multiple programs may run simultaneously, the O.S allocates resources such as CPU time, memory, storage and I/O devices among them.

7. Accounting:- The O.S keeps track of resource usage by users and programs. This can be used for monitoring performance analysis or billing.

8. protection and security:-

The operating system controls access to system resources and prevents unauthorized users or programs from accessing data and hardware.

operating system



provide system services



use hardware



use hardware.

2. Explain the different types of operating systems (Batch, multiprogramming, time sharing, Real time, Distributed, Multi-processor with examples.

→ 1. Batch operating system:-

A Batch operating system collects similar jobs and execute them in groups or batches without requiring user interaction during execution.

features:-

- job are processed one after another.
- No direct interaction b/w the user and the program while it is running.

ex:-

- IBM mainframe system using OS 360

2) Multiprogramming operating system:-

A multiprogramming O.S keeps multiple programs in main memory at the same time. when one program is waiting for I/O, the CPU can execute another program.

Ex: UNIX and early mainframe operating system.

3) Time sharing operating system:-

A time sharing O.S allows multiple users or programs to interact with the computer simultaneously. The CPU gives each process a small amount of time called a time slice.

- provides quick response to users.
- supports multiple users simultaneously.

ex:- UNIX / LINUX, windows and macOS multiple users working on the same server at the same time.

4) Real-time operating system:-

A Real time operating system is designed to respond to events within a specified time limit correctness depends not only on the results but also on completing the task on time.

- There are two main types.

- 1) Hard real time
- 2) Soft real time.

Ex: Vx works, QNX, FreeRTOS

5) Distributed operating system :-

A distributed O.S manages a group of connected computers and makes them appear to users as a single integrated systems.

- Resources can be share on multiple computers.
- Can improve reliability and performance
- Computers communicate through a network.

Ex: Amoeba and plan 9.

- Several networked computers working ~~etc~~ together to a comp

6) Multiprocessor operating system :-

A multiprocessor supports a computer system having two or more processors CPU cores that work concurrently.

- Multiple processors can execute tasks.
- Supports parallel processing.

ex: Linux, Windows, UNIX servers, system running on multiprocessor hardware.

- A server using multiple CPU comes to handle many users and applications simultaneously.

3. Explain real-time operating systems in details differentiate between Hard real-time and soft real-time systems.

→ A Real-time operating system is an operating system designed to process data and respond to events within a specified time limit or deadline.

- In normal operating system getting the correct result is usually the main concern. In a RTOS, both the correctness of the result and the time at which the result is produced are important.

Characteristics of RTOS :-

1) Deterministic Response.

2) Fast interrupt handling

3) priority based scheduling

4) Low Latency.

6) Deadline based operation.

Hard Real time System

soft Real Time system.

① In this system response time is in milliseconds.

① In this system response time are higher

② In this system safety is critical

② Safety is not critical.

③ A hard real time system is

③ A soft real time system is less restriction

④ e.g satellite launch, railway signaling system

④ e.g DVD player, telephone switches, electronic game etc.

⑤ Guarantees response within a specific deadline.

⑤ Does not guarantee response within a specific deadline.

⑥ Focused on processing critical t

4. What are system calls? Explain the different categories of system calls with suitable example.

→ System calls :-

A system call is a mechanism through which a user program requests a service from the operating system (OS) kernel.

- Applications cannot directly access critical resources such as the CPU, memory, files or I/O devices. Instead, they use system calls to ask the OS to perform these operations safely.

- System calls provide the interface between a running program and the OS.

* Categories :-

1) process control :-

These system calls are used to create, execute and terminate processes.

- Create a process.

- load and execute a program

- terminate a process.

- Allocate or release memory.

Ex:-

fork() - creates a new process.

exec() - loads and executes new program.

exit() - terminate.

2) File Management :-

These system calls allow programs to

- Create a file

- Open and close a file.

- Read from a file

- Write to a file.

- Delete a file

- Change a file attribute.

Ex: open(), read(), write(), close(), unlink().

3) Device Management :-

These device management system call is used to request and control I/O devices such as keyboards, printers, disks and displays.

- Request a device
- Release a device
- Read from a device
- Write to a device.
- Get modify device attributes.

Ex:-

- read ()
- write ()
- ioctl () :- a program may use ioctl () to send control commands to a device driver.

4) Information Maintenance :-

These system calls are used to obtain or modify the information about the system and processes.

Ex:-

- getpid () - obtain the process ID.
- time () - obtain the current time
- uname () - obtains information about the operating system.
- getuid () - obtains the user ID.

5) Communication :-

These system calls allow processes to exchange information each other.

ex:-

- pipe () - Creates a communication channel between process.
- socket () - creates network
- send () - sends data
- recv () - receives data.

6) protection and security -

OR among these system calls control access to files, processes and other system resources.

Ex:-

- chmod () - changes file permissions
- chown () - changes file ownership.

- access () - checks accessibility of a file.

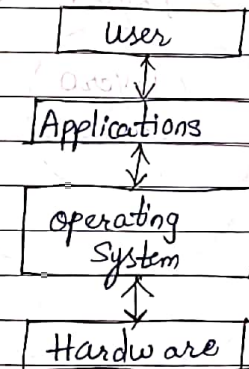
- setuid () - changes the user identity associated with a process.

Q:5. Explain the layered structure or [Simple structure, modular structure] of an operating system.

→ 1) Simple Structure:-

Simple structure operating system have no well-defined architecture and are small, simple and limited in a functionality.

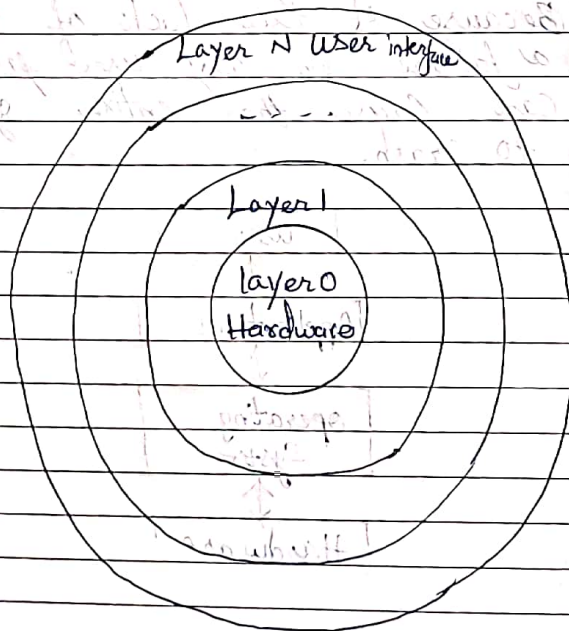
Their interfaces and functional levels are not clearly separated, which makes them less reliable. Because of this lack of separation a failure in any user program can cause the entire system to crash.



2. Layered structure:-

In layered structure the O.S is divided into a number of layers. Each layer performs a specific set of functions.

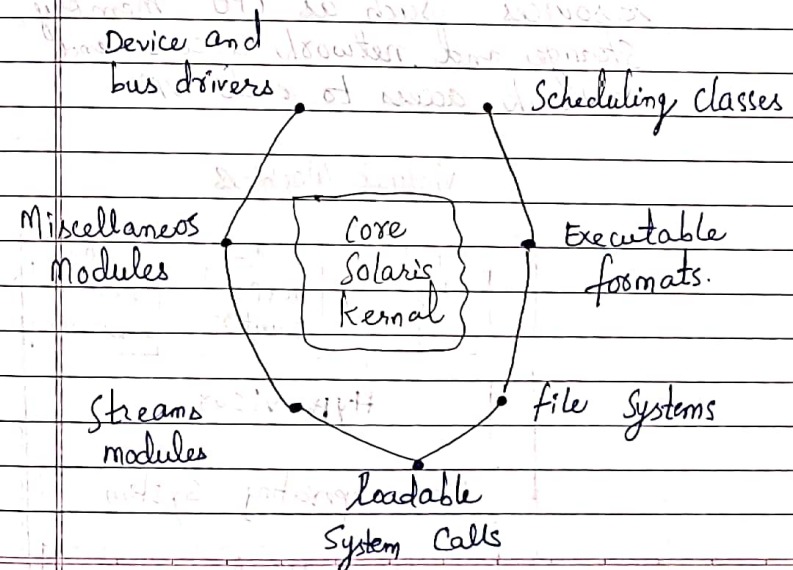
- The lowest layer interacts directly with the hardware, while the highest layer provides services to users and applications.



37. Modular structure:-

A modular structure uses a kernel that contains only essential core components, while additional services are added as loadable modules during boot or runtime.

- Kernel contains only core components.
- Extra services are added as dynamically loadable modules.
- Used in systems like solaris.

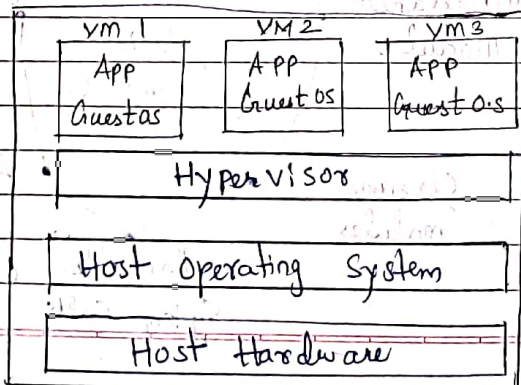


6. Explain Virtual Machine Concept and its advantages.

→ A virtual machines [vm] is a software based computer that behaves like a physical computer. It provides a virtual environment in which an operating system and its applications can run as if they were running on a real machine.

It special software layer called a virtual machine monitor (VMM) or Hypervisor creates and manages virtual machines. it allows physical resources such as CPU memory, storage and network access and network access to each VM.

Virtual Machines



Advantages of virtual Machines:-

- 1) Resource sharing
- 2) Isolation.
- 3) Run multiple operating systems.
- 4) Cost Reduction
- 5) Easy testing and development.
- 6) Backup and Recovery
- 7) Security
- 8) Scalability and flexibility.

Q:7) Discuss System Design and implementation goals of an operating system.

→ The design and implementation of an operating system (OS) involve deciding how the O.S should be structured and then developing it so that it works efficiently, reliably and securely.

- The main goals can be divided into system design goals and implementation goals.

1) System design goals :-

System design goals describe what the operating system should achieve.

① Convenience -

The OS should make the computer easy and convenient to use.

② Efficiency :-

The O.S should use computer resources efficiently

- It should maximize the use of CPU, main memory, storage, I/O devices.

③ Reliability :-

The O.S should operate correctly and consistently. It should detect errors and prevent one faulty program from bringing down the entire system.

④ Security and protection :-

The O.S must protect data and resources from unauthorized access.

Ex:-

- User authentication.
- file permission.
- process isolation
- Access control.

⑤ Stability :-

The O.S should continue to perform well when the system grows.

⑥ portability:-

The O.S should be capable of being adapted to different hardware platform with minimal changes.

⑦ Maintainability:-

The O.S should be designed so that it is easy to modify, update, debug and maintain.

2) System Implementation Goals:-

- Implementation goals describe how the O.S should be developed to achieve the design objectives.

① Modularity:-

The O.S should be divided into separate modules like:

- process management.
- memory management.
- file management.
- I/O management.
- Security.

② Correctness:-

The implementation must perform its intended functions correctly and handle exceptional situation properly.

③ performance:-

The implementation must provide performance with

- low response time
- High throughput.
- Efficient resource utilization.
- low overhead.

④ Flexibility :-

The O.S should be flexible enough to support new hardware, application and technologies without requiring a complete redesign.

⑤ Security :-

Security mechanisms should be incorporated into the implementation rather than added only after the O.S is completed.

⑥ Ease of debugging :-

The O.S should provide mechanisms that help developers to identify and fix errors efficiently.

8. Explain system Generation (SYSGEN) in operating systems.

→ system generation (SYSGEN) is the process of configuring and building an operating system for a particular computer system. It allows the O.S to be customized according to the available hardware and the requirements of the user or organization.

- A computer system may have different types of CPU's, memory size, storage devices, printers, network cards, SYSGEN identifies these components and creates an appropriate OS configuration.

• Why it is needed :-

- An operating system needs information about the hardware on which it will run

- Types and number of processors.
- Amount of main memory.
- Available storage device.
- I/O devices
- Device Controllers.
- Network devices
- Required device drivers.

* Steps in SYSGEN

1) Identify Hardware

- The hardware configuration of the computer is determined.

2) Select required OS components.

- Only the necessary OS components and service may be selected.

3) Configure the kernel:

- The OS kernel is configured according to the selected hardware and software requirements.

4) Compile and link.

The required OS source modules are compiled and linked to create the operating system kernel or system image.

5) Install the generated system.

- The generated OS image is installed on the computer's storage device.

6) Boot the system :-

When the computer starts, the boot loader loads the generated OS kernel into memory and the OS begins execution.

9) Define operating system. Explain its functions and list any four types of OS with examples.

→ An operating system is a system software that acts as an interface between the user and computer hardware and software resources and provides services for application programs.

ex: Windows, Linux, macOS, Android

* Functions :-

1) Process Management :-

Creates, schedules, and terminates process and manages CPU allocation.

2) Memory Management -

Allocates and deallocates main memory to programs and keeps track of memory usage.

3) File Management -

Creates, deletes, reads, writes and organizes file and directories.

4) Device Management -

Controls I/O devices such as keyboard, printers, disks and displays through device drivers.

5) Security and protection -

Prevents unauthorized access to system resources and protects user data.

6) User Interface -

Provides for users to interact with the computer such as a command line interface.

7) Error Detection and Handling -

Detects hardware and software errors and takes appropriate corrective actions.

8) Resource Allocation -

Allocates CPU, memory, storage and other resources among different programs and users.

Types of O.S :-

1) Batch OS :-

executes job in groups without direct user interaction
e.g IBM OS/360

2) Time Sharing OS :-

Allow multiple users
Ex: UNIX / LINUX.

3) Real Time OS :-

provides response within specific time limits.

Ex: Vxworks, FreeRTOS

4) Distributed OS:-

manages multiple computers
as an integrated system

ex! Amoeba